

ADSprofiling&analysis

```
from google.colab import drive
drive.mount('/content/drive')

%cd /content/drive/MyDrive/ds202/project
```

```
Mounted at /content/drive
/content/drive/MyDrive/ds202/project
```

```

import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

train = pd.read_csv("data/prepared/train_full_cor.csv", nrows=1000) # load first 1000 rows
train.head()

```

	SK_ID_CURR	ap_PCT_CHILDREN	ap_PAMT_INCOME_TOTAL	ap_PAMT_CREDIT	ap_PAMT_ANNUITY	ap_PAMT_GOODS_PRICE	ap_REGISTRATION_AREA	ap_P_DAYS_RENT		card_ME_CONTRACT_STATUS_Complicated_mortgage	card_ME_CONTRACT_STATUS_Complicated_mortgage	card_ME_CONTRACT_STATUS_Refused_mortgage	card_ME_CONTRACT_STATUS_Refused_mortgage	card_ME_CONTRACT_STATUS_Declined_mortgage	card_ME_CONTRACT_STATUS_Declined_mortgage	card_ME_CONTRACT_STATUS_Sent_proposal_mortgage	card_ME_CONTRACT_STATUS_Sent_proposal_mortgage	card_ME_CONTRACT_STATUS_Signed_mortgage	card_ME_CONTRACT_STATUS_Signed_mortgage
0	1000002	0	12.218500	12.915581	10.114619	12.768544	0.018801	315.0	21.0	12.2.0	False	False	False	False	False	False	False	False	False
1	1000003	0	12.506181	14.072865	10.482892	13.937287	0.003541	559.0	40.0	40.0	False	False	False	False	False	False	False	False	False
2	1000004	0	11.119898	11.813037	8.817446	11.813037	0.010032	635.0	8.0	14.2.0	False	False	False	False	False	False	False	False	False
3	1000006	0	11.813037	12.652947	10.298481	12.601491	0.008019	634.0	10.1.0	328.0	False	False	False	False	False	False	False	False	False
4	1000007	0	11.707678	13.148033	9.992711	13.148033	0.028663	664.0	10.1.0	144.0	False	False	False	False	False	False	False	False	False

5 rows × 1809 columns

```
# Load application_train.csv for TARGET
target_df = pd.read_csv("input/application_train.csv", usecols=["SK_ID_CURR", "TARGET"])

# Load partial merged data (exclude TARGET)
train = pd.read_csv("data/prepared/train_full_cor.csv", usecols=[
    "SK_ID_CURR", "app_DAYS_BIRTH", "app_AMT_INCOME_TOTAL"
], nrows=100000)

# Merge in TARGET
train = pd.merge(train, target_df, on="SK_ID_CURR", how="left")

# Convert to age
train["AGE"] = (-train["app_DAYS_BIRTH"] / 365).astype(int)
train.rename(columns={"app_AMT_INCOME_TOTAL": "INCOME", "TARGET": "LABEL"}, inplace=True)
```

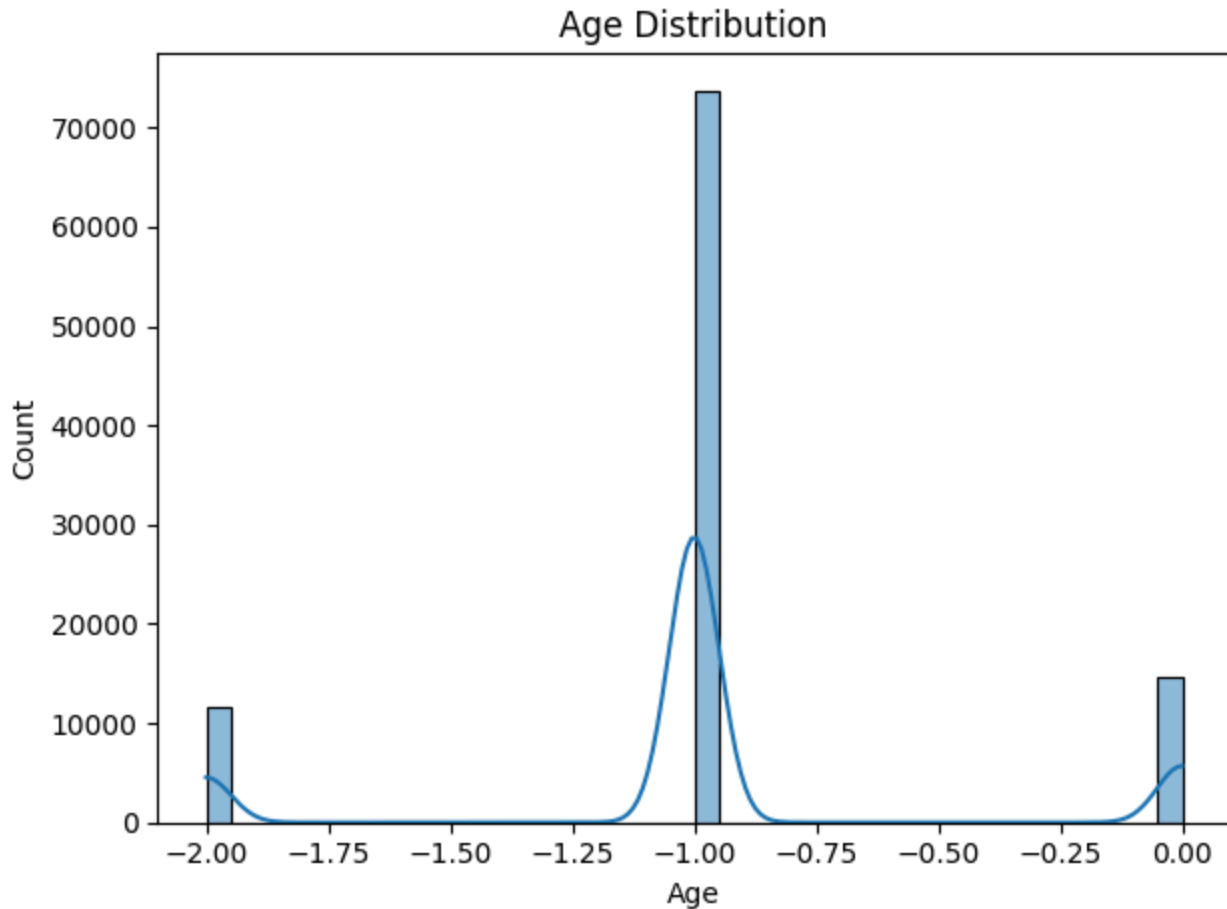
```

# AGE distribution
sns.histplot(train["AGE"], bins=40, kde=True)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.tight_layout()
plt.show()

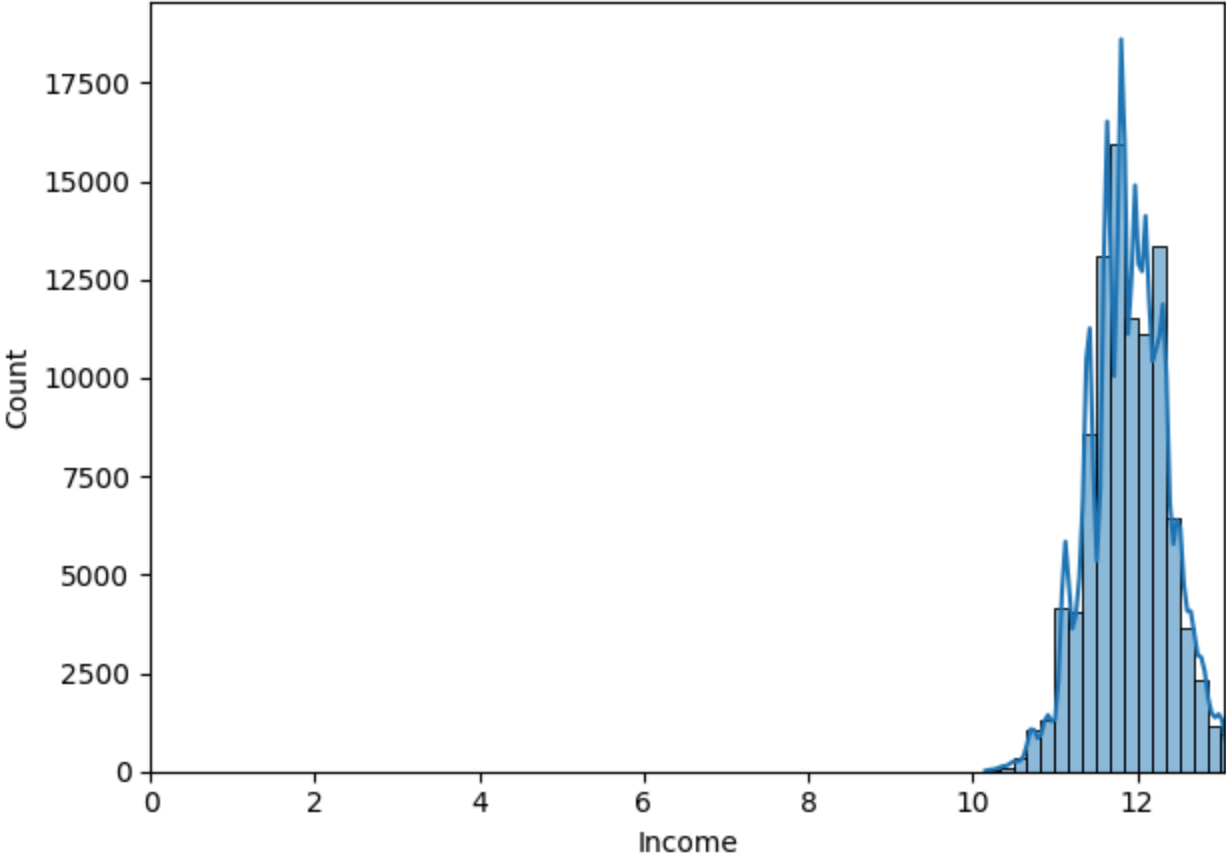
# INCOME distribution (trimmed for outliers)
sns.histplot(train["INCOME"], bins=50, kde=True)
plt.title("Income Distribution")
plt.xlabel("Income")
plt.xlim(0, train["INCOME"].quantile(0.99))
plt.tight_layout()
plt.show()

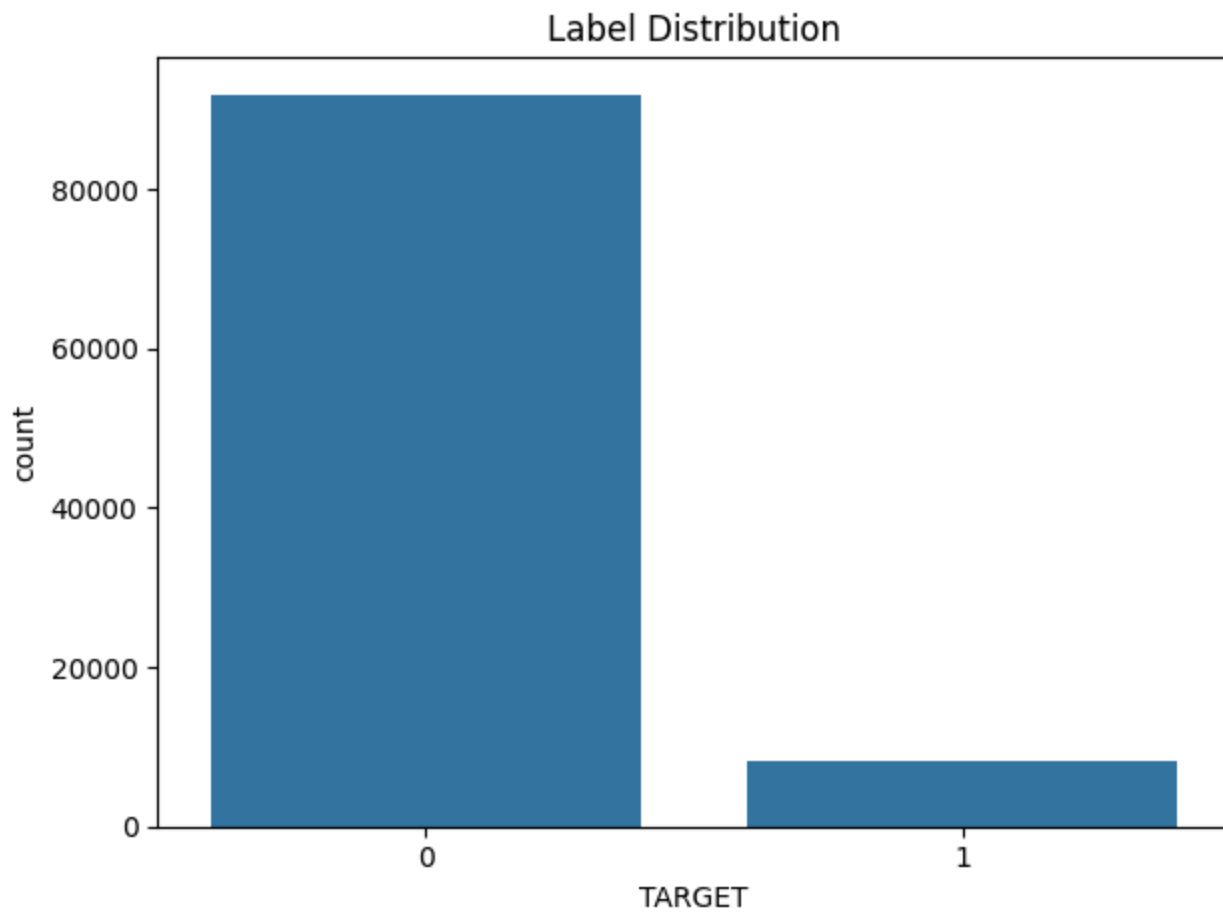
# LABEL distribution (0 = good, 1 = default)
sns.countplot(x="LABEL", data=train)
plt.title("Label Distribution")
plt.xlabel("TARGET")
plt.tight_layout()
plt.show()

```



Income Distribution





```
print("Correlation between AGE, INCOME, and LABEL:")  
print(train[["AGE", "INCOME", "LABEL"]].corr())
```

Correlation between AGE, INCOME, and LABEL:

	AGE	INCOME	LABEL
AGE	1.000000	0.073859	0.062354
INCOME	0.073859	1.000000	-0.020355
LABEL	0.062354	-0.020355	1.000000

```
# AGE groups
age_bins = pd.cut(train["AGE"], bins=[20,30,40,50,60,70], right=False)
print("\nAge Group Frequencies:")
print(age_bins.value_counts().sort_index())

# INCOME bands (quintiles)
income_bins = pd.qcut(train["INCOME"], q=5, duplicates="drop")
print("\nIncome Band Frequencies:")
print(income_bins.value_counts().sort_index())
```

```
Age Group Frequencies:
AGE
[20, 30)    0
[30, 40)    0
[40, 50)    0
[50, 60)    0
[60, 70)    0
Name: count, dtype: int64

Income Band Frequencies:
INCOME
(10.151, 11.503]    20750
(11.503, 11.813]    27908
(11.813, 11.995]    11544
(11.995, 12.324]    24470
(12.324, 18.578]    15328
Name: count, dtype: int64
```

```
print("\nSubset shape:", train.shape)
print("Columns used:", train.columns.tolist())
```

```
Subset shape: (100000, 5)
Columns used: ['SK_ID_CURR', 'INCOME', 'app_DAYS_BIRTH', 'LABEL', 'AGE']
```